



Connect Data Stored in an InterSystems Product to Power BI

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Connect Data Stored in an InterSystems Product to Power BI

This page describes how to access data stored in an InterSystems® product in Power BI using the InterSystems® Health Insight Connector.

1 Introduction to the Connector

The InterSystems Health Insight Connector for Power BI is a custom connector for InterSystems products. It allows you to use Microsoft Power BI to access and create reports on regular relational tables as well as InterSystems IRIS Business Intelligence cube data. It includes full DirectQuery support when querying either type of data. The connector is included with Power BI Desktop.

Important: Prior to July 2024, the InterSystems Health Insight Connector was distributed as the **InterSystems IRIS (Beta)** connector. If your version of Power BI Desktop lists both connectors as options in the **Get Data** menu, select the newer connector (**InterSystems Health Insight**).

To continue using a report or dashboard which was created using the beta connector, you must update it to use the InterSystems Health Insight Connector instead. See [Cannot Connect to Data Source for Existing Report or Dashboard After Update](#).

2 Connect to InterSystems IRIS

Prior to connecting to InterSystems IRIS from Power BI Desktop, ensure that you have an InterSystems IRIS ODBC driver installed on your system.

Note the following:

- The Health Insight Connector does not require you to set up a DSN for operation; only the driver itself is necessary.
- Ensure that you have an InterSystems IRIS ODBC35 driver installed. If you are unsure of which driver to download, please contact the InterSystems Worldwide Response Center (WRC).

In order to connect to InterSystems IRIS from Power BI Desktop, do the following:

1. Open Power BI Desktop and click **Get Data** > **More...** > **InterSystems Health Insight**.
2. Select **Connect**.
3. Enter connection information for your InterSystems IRIS instance. Here, **Host (IP Address)** is the IP address of the host for your InterSystems IRIS instance, **Port** is the instance's superserver port, and **Namespace** is the namespace where your Business Intelligence data is located. Accept all other options as default.
4. Upon your first connection to an instance of InterSystems IRIS, an authentication dialog will appear. Choose **Basic** and enter your InterSystems IRIS credentials.

3 Browse Your Data

If you have successfully connected to InterSystems IRIS, Power BI will display the database **Navigator** dialog. You can browse relational tables by selecting **Tables**. You can expand packages in the left pane to select tables and/or views that you want to include in your Power BI report.

Alternatively, you can view available InterSystems IRIS BI cubes by selecting **Cubes** in the left pane. Expanding the **Cubes** option lists all available InterSystems IRIS Business Intelligence cubes in the current namespace. Note that cubes or subject areas with certain features that cannot be supported through SQL access, such as programmatic filters, are excluded from the list.

When you expand a cube, you will see the star schema representation of the cube, including regular dimensions and a fact table with all regular measures for the cube. Note that some columns with internal identifiers are removed.

4 Publish Reports and Dashboards Using Your Data

Using the Power BI cloud service, you can share reports and dashboards which incorporate data from your InterSystems IRIS cubes and tables. To do so, install and configure a data gateway according to the instructions provided by [the Microsoft documentation](#). Your data gateway and your data sources (including InterSystems IRIS) must be registered on the **Manage connections and gateways** page of the Power BI service.

After you **Publish** your report or dashboard using Power BI Desktop, access the **Gateway connection** settings for the associated dataset in the Power BI service and manually add a mapping to your InterSystems IRIS data source.

5 Troubleshoot the Connector

This section provides guidance regarding some common problems you may encounter when using the InterSystems Health Insight Connector for Power BI.

5.1 Cannot Connect to Data Source for Existing Report or Dashboard After Update

If you have created a report or dashboard which used the **InterSystems IRIS (Beta)** connector to access data on an InterSystems IRIS instance, attempts to access that report or dashboard after updating to the July 2024 release of Power BI will yield an error. This is because the beta connector has been removed, so Power BI will be unable to load the model for the report or dashboard because it will be unable to connect to your InterSystems IRIS data source.

To restore access to a report or dashboard that you created using the beta connector, update it to use the InterSystems Health Insight Connector by performing the following steps:

1. Add the data source using the InterSystems Health Insight Connector by following the instructions described in [Connect to InterSystems IRIS](#).
2. For each query in the report or dashboard:
 - a. Update the query's source so that it is `InterSystemsHealthInsight.Database` instead of `IRIS.Database`.

Note: Source names are case sensitive.

- b. As needed, update the connector's connection detail parameters, and provide connection credentials.

For guidance editing queries, refer to the Microsoft Power BI documentation: <https://learn.microsoft.com/en-us/power-bi/transform-model/desktop-query-overview>.

5.2 Missing Tables in the Navigator

The InterSystems Health Insight Connector for Power BI excludes system tables and tables associated with InterSystems IRIS Business Intelligence cubes from the regular **Tables** menu. Scrubbed and annotated versions of the latter are available through the **Cubes** menu. If you need access to a table or a field not listed in the Navigator, you can add it manually with a [custom query](#) or use Power BI's generic ODBC connector.

5.3 Missing Cubes in the Navigator

The InterSystems Health Insight Connector for Power BI leverages the relational projects of InterSystems Business Intelligence cubes to make them available for use in Power BI. Some cube features, like programmatic filters, cannot be supported through these projections and are therefore left out of the list. Please contact the WRC if you encounter a cube where this behavior is not appropriate.

5.4 Dimension Hierarchy Not Appearing in the Report Designer

Power BI does not currently allow seeding dimension information from a connector.

5.5 Multilevel Dimension Hierarchy Not Functioning Correctly

When a dimension has multiple levels, these levels are usually represented by separate dimension tables (snowflake schema). While foreign key relationships exist between the fact table and each dimension level and between the different levels of the dimension, Power BI can only choose one path from a fact table to a higher dimension level as the “active relationship”, and may choose the wrong one, leading to unexpected query results. To fix the active relationship, click **Manage Relationships** in Power BI Desktop and de-activate the direct links between a fact table and higher-level dimension tables. Then, activate the correct relationships one by one. For more information, see the Microsoft [documentation](#).

5.6 Access Denied Error Appears When You Attempt to Connect to InterSystems IRIS

To connect to InterSystems IRIS from Power BI Desktop, you must have EXECUTE privileges on the following stored procedures:

- %DeepSee_SQL.GetCubes
- %DeepSee_SQL.GetDimensionTables
- %DeepSee_SQL.GetDimensionColumns
- %DeepSee_SQL.GetUnsupportedFeatures

An administrator can use the [GRANT](#) command to grant privileges to you.

5.7 Power BI Service Cannot Retrieve Data for a Report or Dashboard

At this time, to populate a published report or dashboard with data from the InterSystems Health Insight Connector for Power BI, you must manually create a mapping between the associated dataset and your InterSystems IRIS data source, as

described in [Publish Reports and Dashboards Using Your Data](#). Note that if you have already registered InterSystems IRIS as a data source within the Power BI service, this action may create a duplicate entry for your InterSystems IRIS data source. After you have successfully established access to the InterSystems IRIS data included within the dataset, you can specify either entry and then remove the duplicate.